

```
import math
def brahian_al_cuadrado(x):
    y=x**2
    return y
```

```
brahian_al_cuadrado(2)
```

```
4
```

```
def brahian_1(x):
    y=math.pi*x+1
    return y
brahian_1(2)
```

```
7.283185307179586
```

```
def brahian_3(x):
    y=((x+3)/((x**2)+1))
    return y
brahian_3(5)
```

```
0.3076923076923077
```

```
def brahian4(x):
    y=math.log10((x+3*x**2)/(2+5*math.pi))
    return y
brahian4(1)
```

```
-0.6461086211123684
```

```
def brahian5(x):
    num=(5*x+4)
    dem=(math.pi+7*x)
    y=math.exp(num/dem)
    return y
brahian5(1)
```

```
2.4288905233001983
```

```
x1=[10,2,3,40,5]
print(x1)
print(x1[0])
print(x1[1])
```

```
[10, 2, 3, 40, 5]
10
2
```

```
for x in x1:
    # Bloque de código a ejecutar por cada elemento}
```

```
print(x)
```

```
10  
2  
3  
40  
5
```

```
for x in x1:  
    print(brahian4(x))  
    print(x)
```

```
1.2431930813939418  
10  
-0.10204057676209283  
2  
0.2289526422793316  
3  
2.4366767492040817  
40  
0.6549213745516128  
5
```

```
for x in range(5):  
    print(x)
```

```
0  
1  
2  
3  
4
```

```
for x in range(5):  
    print(brahian_3(x))
```

```
3.0  
2.0  
1.0  
0.6  
0.4117647058823529
```

```
for x in range(-5,5):  
    print(brahian_1(x))  
    print(x)
```

```
-14.707963267948966  
-5  
-11.566370614359172  
-4  
-8.42477796076938  
-3  
-5.283185307179586  
-2  
-2.141592653589793  
-1  
1.0
```

```
0
4.141592653589793
1
7.283185307179586
2
10.42477796076938
3
13.566370614359172
4
```

```
abs(-1)
```

```
1
```

```
def brahian_valor_abosluto(x):
    x=1
    if x>0:
        x
    elif x<0:
        y=-x
    else:x
    y=0
    return y
brahian_valor_abosluto(-1)
```

```
0
```

```
for x in range(-5,5):
    print(brahian_valor_abosluto(x))
```

```
0
0
0
0
0
0
0
0
0
0
0
```

```
def numero_mayor(a,b,c):
    if a>b and a>c:
        y=a
    elif b>c and b>a:
        y=b
    else:
        y=c
    return y

numero_mayor(-1,-3,-5)
```

```
-1
```

```
# Creación de un arreglo (lista) de frutas
frutas = ["manzana", "banana", "cereza", "naranja"]

# Acceder a elementos
print(frutas[0])      # Imprime: manzana (primer elemento)
print(frutas[-1])     # Imprime: naranja (último elemento)
print(frutas[1:3])    # Imprime: ['banana', 'cereza'] (slicing/segmento)

# Modificar un elemento
frutas[1] = "mango"

# Agregar elementos
frutas.append("uva")  # Agrega al final
frutas.insert(2, "pera") # Inserta en la posición 2

# Eliminar elementos
frutas.remove("cereza") # Elimina por valor
ultima = frutas.pop()   # Elimina y devuelve el último

print(frutas) # Imprime: ['manzana', 'mango', 'pera', 'naranja']
```

```
manzana
naranja
['banana', 'cereza']
['manzana', 'mango', 'pera', 'naranja']
```